LN2 General Probability Theory

Kiseop Lee

Department of Statistics
Purdue University

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Outlines

Textbook: Stochastic Calculus for Finance II: Continuous-Time Models: Chapter 1

1. Infinite Probability Space
2. Random Variables and Distributions
3. Expectations
4. Convergence of Integrals
5. Computation of Expectation
6. Change of measure

σ-algebra

Definition 1

Let Ω be a non-empty set. $\mathcal{F}$ is a collection of subsets of Ω. We say that $\mathcal{F}$ is a σ-algebra provided that

(1) $\Phi \in \mathcal{F}$

(2) $A \in \mathcal{F} \Longrightarrow A^C \in \mathcal{F}$

(3) $A_n \in \mathcal{F}, \forall n \Longrightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

A Probability Space

Definition 2

Let Ω be a non-empty set. $\mathcal{F}$ is a σ-algebra of Ω. A probability measure $\mathbb{P}$ is a function that, to every set $A \in \mathcal{F}$, assigns a number in $[0,1]$, called the probability of A and written $\mathbb{P}(A)$. We require:

(1) $\mathbb{P}(\Omega) = 1$

(2) *For $A_1, A_2, A_3, \cdots$, $A_i \cap A_j = \Phi$ if $i \neq j \Longrightarrow$* $\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$

The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

Example 1: Uniform (Lebesgue) measure on [0,1]

We construct a mathematical model for choosing a number at random from the unit interval $[0, 1]$ so that the probability is distributed uniformly over the interval.

- $[a, b]$ is a closed interval on $[0, 1]$.
- $\mathbb{P}[a, b] = b - a, 0 \leq a \leq b \leq 1$. (Lebesgue measure, denoted as $\mathcal{L}$)
- $(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}]$
- The σ-algebra obtained by beginning with closed intervals and adding everything else necessary in order to have a σ-algebra is called the *Borel σ-algebra* of subsets of [0,1] and is denoted $\mathcal{B}[0,1]$. The sets in this σ-algebra are called *Borel sets*. Every subsets of [0,1] we encounter in this text is a Borel set.

Example 2: Infinite, independent coin-toss space

We toss a coin infinitely many times and let Ω_∞ denote the set of possible outcomes. We assume the probability of head on each toss is $p > 0$, the probability of tail is $q = 1 - p > 0$, and the different tosses are independent. We want to construct a probability measure corresponding to this random experiment.

- $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1 \Rightarrow \mathcal{F}_0 = \{\emptyset, \Omega\}$
- $A_H =$ the set of all sequences beginning with $H = \{\omega; \omega_1 = H\}$
  $A_T =$ the set of all sequences beginning with $T = \{\omega; \omega_1 = T\}$
  $\mathbb{P}(A_H) = p$ and $\mathbb{P}(A_T) = q \Rightarrow \mathcal{F}_1 = \{\emptyset, \Omega, A_H, A_T\}$
- $\mathcal{F}_2 = \{\emptyset, \Omega, A_H, A_T, A_{HH}, A_{HT}, A_{TH}, A_{TT}, A_{HH}^c, A_{HT}^c, A_{TH}^c, A_{TT}^c,$
  $A_{HH} \cup A_{TH}, A_{HH} \cup A_{TT}, A_{HT} \cup A_{TH}, A_{HT} \cup A_{TT}\}$
- ...
- $\mathcal{F}_\infty$

Almost Sure

Definition 3

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If a set $A \in \mathcal{F}$ satisfies $\mathbb{P}(A) = 1$, we say that the event A occurs almost surely.

Random Variable

Definition 4

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable is a real-valued function X defined on Ω with the property that for every Borel subset B of $\mathbb{R}$, the subset of Ω given by $\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}$ is in the σ-algebra $\mathcal{F}$.

Example 3: Stock prices

Continuation of Example 2. Recall the independent, infinite coin-toss space $(\Omega_\infty, \mathcal{F}_\infty, \mathbb{P})$. Let us define stock prices by the formulas

$$S_0(\omega) = 4 \text{ for all } \omega \in \Omega_\infty,$$

$$S_1(\omega) = \begin{cases} 8 \text{ if } \omega_1 = H, \\ 2 \text{ if } \omega_1 = T, \end{cases}$$

$$S_2(\omega) = \begin{cases} 16 \text{ if } \omega_1 = \omega_2 = H, \\ 4 \text{ if } \omega_1 \neq \omega_2, \\ 1 \text{ if } \omega_1 = \omega_2 = T, \end{cases}$$

and, in general,

$$S_{n+1}(\omega) = \begin{cases} 2S_n(\omega) \text{ if } \omega_{n+1} = H, \\ \frac{1}{2}S_n(\omega) \text{ if } \omega_{n+1} = T. \end{cases}$$

Example 3: Stock prices (continued)

- All these are random variables.
- They assign a numerical value to each possible sequence of coin tosses.
- We can compute the probabilities that these random variables take various values.
- For example, $\mathbb{P}\{S_2 = 4\} = \mathbb{P}(A_{HT} \cup A_{TH}) = 2pq$.

Distribution

Definition 5

Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. The distribution measure of X is the probability measure μ_X that assigns to each Borel subset B of $\mathbb{R}$ the mass $\mu_X(B) = \mathbb{P}\{X \in B\}$.

- Definition 5 for the distribution measure of a random variable makes sense for both discrete random variables and continuous variables.
- Random variables have distributions, but distributions and random variables are different concepts.
  - Two different random variables can have the same distribution.
  - A single random variable can have two different distributions.

Example 4: Uniform measure

Let $\mathbb{P}$ be the uniform measure on $[0,1]$. Define $X(\omega) = \omega$ and $Y(\omega) = 1 - \omega$ for all $\omega \in [0,1]$. Then the distribution measure of X is uniform, i.e.,

$$\mu_X[a,b] = \mathbb{P}\{\omega : a \leq X(\omega) \leq b\} = \mathbb{P}[a,b] = b - a, 0 \leq a \leq b \leq 1$$

Although the random variable Y is different from the random variable X, Y has the same distribution as X:

$$\mu_Y[a,b] = \mathbb{P}\{\omega : a \leq Y(\omega) \leq b\} = \mathbb{P}[1-b, 1-a] = b - a = \mu_X[a,b]$$

Now suppose we define another probability measure $\tilde{\mathbb{P}}$ on $[0,1]$ by specifying

$$\tilde{\mathbb{P}}[a,b] = \int_a^b 2\omega d\omega = b^2 - a^2, 0 \leq a \leq b \leq 1.$$

Now, $\tilde{\mu}_X[a,b] \neq \tilde{\mu}_Y[a,b]$

Expectation

Review of Integrals.

Definition 6

Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. The expectation of X is defined to be

$$\mathbb{E}X = \int_{\Omega} X(\omega) d\mathbb{P}(\omega).$$

Properties

Theorem 1

Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

(1) *If Ω is finite, then*

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega)$$

(2) *(Integrability) X is integrable iff*

$$E|X| < \infty.$$

Properties

Theorem 1 Continued

Now let Y be another random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

(3) (Comparison) If $X \leq Y$ almost surely and X and Y are integrable, then

$$\mathbb{E}X \leq \mathbb{E}Y.$$

(4) (Linearity)

$$\mathbb{E}(\alpha X + \beta Y) = \alpha \mathbb{E}X + \beta \mathbb{E}Y.$$

(5) (Jensen's inequality) If ϕ is a convex, real-valued function, and $E|X| < \infty$, then

$$\phi(\mathbb{E}X) \leq \mathbb{E}\phi(X).$$

Example 5: Infinite coin toss

Let

$$Y_n(\omega) = \begin{cases} 1 & \text{if} \quad \omega_n = H \\ 0 & \text{if} \quad \omega_n = T \end{cases}.$$

Then $\mathbb{E}Y_n = 1 \cdot \mathbb{P}\{Y_n = 1\} + 0 \cdot \mathbb{P}\{Y_n = 0\} = \frac{1}{2}$.

Convergence Almost Surely

Definition 7

Let $X_1, X_2, X_3, \cdots$ be a sequence of random variables, all defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X be another random variable defined on this space. We say that $X_1, X_2, X_3, \cdots$ converges to X almost surely and write

$$\lim_{n\to\infty} X_n = X \text{ almost surely}$$

if the set of $\omega \in \Omega$ for which $X_1(\omega), X_2(\omega), X_3(\omega), \cdots$ has limit $X(\omega)$ is a set with probability one.

Convergence Almost Everywhere

Definition 8

Let $f_1, f_2, f_3, \cdots$ be a sequence of real-valued, Borel-measurable functions defined on $\mathbb{R}$. Let f be another real-valued, Borel-measureable function defined on $\mathbb{R}$. We say that $f_1, f_2, f_3, \cdots$ converges to f almost everywhere and writh

$$\lim_{n\to\infty} f_n = f \text{ almost everywhere}$$

if the set of $x \in \mathbb{R}$ for which $f_1(x), f_2(x), f_3(x), \cdots$ does not have limit $f(x)$ is a set with Lebesgue measure zero.

Convergence of Integrals

Often when random variables converge almost surely, their expected values converge to the expected value of the limiting random variable. Likewise, when functions converge almost everywhere, their Lebesgue integrals converge to the Lebesgue integral of the limiting function. This is not always the case, however. To get the integrals of a sequence of functions to converge to the integral of the limiting function, we need to impose some condition. One condition is given in the following theorem.

Monotone Convergence

Theorem 2

Let $X_1, X_2, X_3, \cdots$ be a sequence of random variable converging almost surely to another random variable X. If

$$0 \leq X_1 \leq X_2 \leq X_3 \leq \cdots \text{ almost surely,}$$

then

$$\lim_{n \to \infty} \mathbb{E}X_n = \mathbb{E}X.$$

Monotone Convergence

Theorem 2 Continued

Let $f_1, f_2, f_3, \cdots$ be a sequence of Borel-measurable functions on $\mathbb{R}$ converging almost everywhere to a function f. If

$$0 \leq f_1 \leq f_2 \leq f_3 \leq \cdots \text{ almost everywhere,}$$

then

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f_n(x)dx = \int_{-\infty}^{\infty} f(x)dx.$$

Dominated Convergence

The following theorem provides another common situation in which the limit of the integrals of a sequence of functions is the integral of the limiting function.

Theorem 3

Let $X_1, X_2, X_3, \cdots$ be a sequence of random variable converging almost surely to a random variable X. If there is another random variable Y such that $\mathbb{E}Y < \infty$ and $|X_n| \leq Y$ almost surely for every n, then

$$\lim_{n\to\infty} \mathbb{E}X_n = \mathbb{E}X.$$

Dominated Convergence

Theorem 3 Continued

Let $f_1, f_2, f_3, \cdots$ be a sequence of Borel-measurable functions on $\mathbb{R}$ converging almost everywhere to a function f. If there is another function g such that $\int_{-\infty}^{\infty} g(x)dx < \infty$ and $|f_n| \leq g$ almost everywhere for every n, then then

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f_n(x)dx = \int_{-\infty}^{\infty} f(x)dx.$$

Computation of Expectation

Let X be a random variable on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We have defined the expectation of X to be the Lebesgue integral

$$\mathbb{E}X = \int_{\Omega} X(\omega) d\mathbb{P}(\omega),$$

the idea being to average the values of $X(\omega)$ over Ω, taking the probabilities into account. But the abstract space Ω is not a pleasant environment in which to actually compute integrals. For computation, we need to rely on densities of the random variables, and we integrate these over the real numbers rather than over Ω. In this section, we develop the relationship between integrals over Ω and integrals over $\mathbb{R}$.

Computation of Expectation

Recall that the distribution measure of X is the probability measure μ_X defined on $\mathbb{R}$ by

$$\mu_X(B) = \mathbb{P}\{X \in B\} \text{ for every Borel subset B of } \mathbb{R}.$$

Theorem 4

Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let g be a Borel-measurable function on $\mathbb{R}$. Then

$$\mathbb{E}|g(X)| = \int_{\mathbb{R}} |g(x)| d\mu_X(x),$$

and if this quantity is finite, then

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x) d\mu_X(x).$$

Computation of Expectation

Theorem 4 tells us that in order to compute the Lebesgue integral $\mathbb{E}g(X) = \int_\Omega g(X(\omega))d\mathbb{P}(\omega)$ over the abstract space Ω, it suffices to compute the integral $\int_{\mathbb{R}} g(x)d\mu_X(x)$ over the set of real numbers.

Computation of Expectation

The common case is when X has a density. This means that there is a nonnegative, Borel-measurable function f defined on $\mathbb{R}$ such that

$$\mu_X(B) = \int_B f(x)dx \text{ for every Borel subset B of } \mathbb{R}.$$

In this case, we can use this density to compute expectations.

Theorem 5

Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let g be a Borel-measurable function on $\mathbb{R}$. Suppose that X has a density f. Then

$$\mathbb{E}|g(X)| = \int_{\mathbb{R}} |g(x)|f(x)dx.$$

If this quantity is finite, then

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x)f(x)dx.$$

Change of measure

In models of financial markets, we need to change from the actual probability $\mathbb{P}$ to the risk-neutral probability measure $\widetilde{\mathbb{P}}$ on a space Ω. We could use a positive random variable Z to accomplish it. Specifically, to change from $\mathbb{P}$ to $\widetilde{\mathbb{P}}$, we need to reassign probabilities in Ω using Z to tell us where in Ω we should revise the probability upward (where $Z > 1$) and where we should revise the probability downward (where $Z < 1$). However, we should do this set-by-set, rather than ω-by-ω. The process is described by the following theorem.

Change of measure

Theorem 6

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let Z be an almost surely nonnegative random variable with $\mathbb{E}Z = 1$. For $A \in \mathcal{F}$, define

$$\widetilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega).$$

Then $\widetilde{\mathbb{P}}$ is a probability measure. Furthermore, if X is a nonnegative random variable, then

$$\widetilde{\mathbb{E}}X = \mathbb{E}[XZ].$$

If Z is almost surely strictly positive, we also have

$$\widetilde{\mathbb{E}}Y = \mathbb{E}\left[\frac{Y}{Z}\right]$$

for every nonnegative random variable Y.

Note that $\widetilde{\mathbb{E}}X = \int_\Omega X(\omega) d\widetilde{\mathbb{P}}(\omega)$.

Change of measure

Remark If X can take both positive and negative values. We may apply the theorem to its positive and negative parts X^+ and X^-, and then subtract the resulting equations, provided the subtraction does not result in an $\infty - \infty$.

Equivalent Probability Measures

Definition 9

Let Ω be a nonempty set and $\mathcal{F}$ a σ-algebra of subsets of Ω. Two probability measures $\mathbb{P}$ and $\widetilde{\mathbb{P}}$ are said to be equivalent if they agree which sets in $\mathcal{F}$ have probability zero.

Under the assumptions of Theorem 6, we can show that $\mathbb{P}$ and $\widetilde{\mathbb{P}}$ are equivalent. See textbook for more details. Because the sets with probability one are complements of the sets with probability zero, $\mathbb{P}$ and $\widetilde{\mathbb{P}}$ agree which sets have probability one as well.

Probability Measures in Financial Models

"In financial models, we will first set up a sample space Ω, which one can regard as the set of possible scenarios for the future. We imagine this set of possible scenarios has an actual probability measure $\mathbb{P}$. However, for purposes of pricing derivative securities, we will use a risk-neutral measure $\widetilde{\mathbb{P}}$. We will insist that these two measures are equivalent. They must agree on what is possible and what is impossible; they may disagree on how probable the possibilities are. After we have $\mathbb{P}$ and $\widetilde{\mathbb{P}}$, we shall determine prices of derivative securities that allow us to set up hedges that work with $\widetilde{\mathbb{P}}$-probability one. These hedges then also work with $\mathbb{P}$-probability one. Although we have used the risk-neutral probability to compute prices, we will have obtained hedges that work with probability one under the actual probability measure."

"A hedge that works almost surely under one assumption of probabilities works almost surely under the other assumption as well, since the probability measures agree which events have probability one."

Change of measure

In general, we may rewrite the equations in Theorem 6 as

$$\int_\Omega X(\omega) d\widetilde{\mathbb{P}}(\omega) = \int_\Omega X(\omega) Z(\omega) d\mathbb{P}(\omega),$$

$$\int_\Omega Y(\omega) d\mathbb{P}(\omega) = \int_\Omega \frac{Y(\omega)}{Z(\omega)} d\widetilde{\mathbb{P}}(\omega).$$

A good way to remember these equations is to write $Z(\omega) = \frac{d\widetilde{\mathbb{P}}(\omega)}{d\mathbb{P}(\omega)}$. Z is somehow a "ratio of probabilities".

Radon-Nikodym derivative

Definition 10

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\widetilde{\mathbb{P}}$ be another probability measure on $(\Omega, \mathcal{F})$ that is equivalent to $\mathbb{P}$, and let Z be an almost surely positive random variable that relates $\mathbb{P}$ and $\widetilde{\mathbb{P}}$ via

$$\widetilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega).$$

Then Z is called the Radon-Nikodym derivative of $\widetilde{\mathbb{P}}$ with respect to $\mathbb{P}$ and we write

$$Z = \frac{d\widetilde{\mathbb{P}}}{d\mathbb{P}}.$$

Radon-Nikodym Theorem

Theorem 7

Let $\mathbb{P}$ and $\widetilde{\mathbb{P}}$ be equivalent probability measures define on $(\Omega, \mathcal{F})$. Then there exists an almost surely positive random variable Z such that $\mathbb{E}Z = 1$ and

$$\widetilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega) \text{ for every } A \in \mathcal{F}$$